



K L Deemed to be University

Department of Electronics and Communication Engineering -- KLVZA

Course Handout

2025-2026, Odd Sem

Course Title	:BASIC ELECTRICAL AND ELECTRONIC CIRCUITS
Course Code	:25EC1101
L-T-P-S Structure	: 3-0-2-0
Pre-requisite	:
Credits	: 4
Course Coordinator	:SRINIVASARAO ALLURI
Team of Instructors	:
Teaching Associates	:

Syllabus :CO-1: AC & DC Fundamentals: DC Fundamentals: Basic circuit elements and sources, Series and Parallel connection of circuit elements, Star-Delta transformation, Ohm's Law, Kirchhoff's Laws. AC fundamentals: Peak Value, Time period, Frequency, RMS & Average values, Form factor, peak factor and power factor. CO-2: Circuit Analysis Techniques: Mesh analysis, Nodal analysis, Thevenin's Theorem, Norton's Theorem, Maximum power transfer theorem, Superposition theorem, Steady-State Analysis of RL, RC, RLC, Resonance and its parameters (Resonant Frequency, Q-Factor, Bandwidth). CO-3: Semiconductor devices: PN Junction diode (Operation, VI Characteristics), Applications: Diode as a switch, Rectifiers (HWR, FWR), Clippers and Clampers, Zener Diode (Breakdown) and voltage regulator, LED Bipolar Junction Transistor: Operation of Transistor, CB, CE, CC Configurations-input and output characteristics and applications, Transistor as switch. CO-4: Integrated Circuits: Operational Amplifiers IC 741 and applications, 555 Timer and applications, Voltage regulators: 78XX, 79XX and LM723.

Text Books :1. Electrical Circuit Theory and Technology, John Bird 6th ed., Routledge publishers Pearson/PHI. 2. Circuits and Networks: Analysis and Synthesis A Sudhakar, Shyam Mohan S Palli, 5th ed, TMH. 3. Electronic Devices and Circuit Theory Robert L. Boylestad 12th ed,Mc Graw Hill. 4. OP-Amps and linear integrated circuit technology by Ramakanth A Gayakwad, 4th ed, Pearson, PHI.

Reference Books :1. Engineering Circuit Analysis, Hayt and Kemmerly, 10th ed., McGraw Hill. 2. A Text Book of Electrical Technology, B.L. Theraja, A.K. Theraja, S. Chand & Company Ltd. 3. Basic Electrical Engineering, V. K. Mehta, Rohit Mehta, S. Chand & Company Ltd. 4. Electronics Devices and Circuits, Jacob Millman & Halkias, 3rd ed., TMH. 5. Electronic Devices and Circuits, David A. Bell, 5th ed., PHI.

Web Links :1. NOC Jan 2020: Network Analysis - YouTube, 2. <https://www.youtube.com/playlist?list=PLBlnK6fEyqRgLR-hMp7wem-bdVN1iEhsh>, 3. <https://www.youtube.com/@Profmad/playlists>

MOOCS :NPTEL/SWAYAM - 1. nptel.ac.in/courses/117106108, 2. nptel.ac.in/courses/108105112 Coursera – 1. <https://www.coursera.org/learn/linear-circuits-dcanalysis>, 2. <https://www.coursera.org/learn/linear-circuits-ac-analysis>, 3. <https://www.coursera.org/learn/electronics> edX – 1. <https://www.edx.org/xseries/mitx-circuits-and-electronics>

Course Rationale :This course is designed for students who are beginning their journey into engineering. By covering both DC and AC circuit fundamentals, along with essential semiconductor devices and integrated circuits, students will gain a solid foundation that can be built upon in more advanced courses. Understanding circuit analysis, along with practical applications of key components like diodes, transistors, and ICs, will equip students to design and troubleshoot a wide range of electrical and electronic circuits. This knowledge is crucial for those aiming to pursue careers in electronics, electrical engineering, or related fields.

Course Objectives :This course is designed to provide students with a foundational understanding of both theoretical and practical aspects of electrical and electronic circuits. The course covers essential topics, ranging

from DC and AC fundamentals to semiconductor devices and integrated circuits, helping students gain the necessary skills for analyzing, and designing electrical & electronic circuits. The course objectives typically include: 1. Gain a solid understanding of DC and AC circuit fundamentals, including basic circuit elements, Ohm's Law, Kirchhoff's Laws, and AC signal parameters. 2. Be proficient in circuit analysis techniques, including mesh and nodal analysis, as well as simplifying circuits using Thevenin's, Norton's, Maximum power transfer and Superposition theorems. 3. Acquire knowledge of semiconductor devices like diodes and transistors, and understand their applications in rectifiers, voltage regulation, and switching. 4. Understand the integrated circuits like operational amplifiers, 555 timers, and voltage regulators in practical applications.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Understand the basic concepts of AC and DC circuits and apply them to solve electric circuit problems.	PSO1,PO1	2
CO2	Apply different circuit analysis techniques to find voltage, current, impedance and power in electrical circuits.	PSO1,PO1	3
CO3	Comprehend the behavior of basic electronic components, such as diodes and transistor, and their applications.	PSO1,PO1	2
CO4	Understand the basic functional principles of different analog ICs, such as IC 741 Op-Amp, IC 555 Timer and IC Voltage Regulators.	PSO1,PO1	2
CO5	Utilize measurement tools and equipment for validating theoretical concepts of different electrical and electronic circuits and develop problem-solving capabilities through hands-on experimentation.	PSO1,PO1,PO5	4

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4
CO1	2	Btl-1 Recall the fundamental concepts of electricity, Time Period, Frequency and Phase of AC signals	Btl-2 Describe the basic Circuit Elements, Ohm's Law, KVL and KCL, Circuit Reduction Techniques: Series, Parallel, and Star-Delta, RMS value, Average value, Form, Peak & Power factors, Reactance, Impedance, Phase in R, L, and C, Real, reactive and apparent powers and power factor		
CO2	3	Btl-1 Identify branch, node,	Btl-2 Explain about Steady	Btl-3 Apply Mesh & Nodal analysis, Thevenin's	

		loop and mesh in a circuit.	state analysis of RL, RC and RLC Circuits	& Norton's theorem, Superposition theorem, Maximum power transfer theorem, Series Resonance technique to complex electric circuits	
CO3	2	Btl-1 List the types of materials and recollect the concepts of semiconductor physics,	Btl-2 Describe operation of the diode, Rectifiers, Clippers, and Clampers. Explain about Zener Diode as a regulator, Operation of LED, Transistor Configurations and transistor as a switch		
CO4	2	Btl-1 List out different types of ICs	Btl-2 Explain about Op-Amp, Inverting and Non-inverting amplifiers, 555 Timer and its modes of operation, IC voltage regulators: 78XX, 79XX and LM723		
CO5	4	Btl-1 Remember the fundamental concepts, terminology, and components used in electrical and electronic circuits laboratory.	Btl-2 Explain and interpret the basic functioning of circuits and systems.	Btl-3 Apply the theoretical concepts to design and implement circuits using hands-on methods and simulation tools and test the performance of circuits.	Btl-4 Analyze different electrical and electronic circuits by conducting hands-on sessions, Analyze practical applications of different analog circuits using simulation tools as well as hardware components. Explore the functionality of voltage limiters, dc level shifters, adders, buffers, waveform generators and traffic light controlling system to understand the real world relevance and versatility of analog electronic systems.

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PSO1	Apply semiconductor physics, circuit analysis, and design techniques in the development of electronics systems.
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-1	Briefing on Course Rationale, Course Handout (CHO) and Evaluation methods	NA	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
2	CO1	COI-2	Basic Definitions, Passive Elements – R, L, and C, Ideal and Practical Sources, Ohm's Law	TBOOK [2], CH1, Page no 1-11	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
3	CO1	COI-2	KCL and KVL, Solving Problems	T BOOK [2], CH1,Page no 11-20	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
4	CO1	COI-2	Series & Parallel Connections of R, L, and C, Current Division and Voltage Division Rules, Source Transformation	T BOOK [2], CH1,Page no 14-22	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
5	CO1	COI-2	Star-to-Delta and Delta-to-Star Transformation and Solving Problems	R BOOK [2],CH2,Page no137-143	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
6	CO1	COI-1	Basic Definitions of AC Signals, Average & RMS Values, Peak & Form Factors of Different AC Signals	R BOOK [2], CH11,Page no 454-464	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
7	CO1	COI-2	Analysis of R, L and C with AC Signal –	T BOOK [2], CH4Page no 174-184.	Chalk,PPT,Talk	ALM,End Semester Exam,Home

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			Impedance, Reactance, Phase and Phase difference			Assignment,SEM-EXAM1
8	CO1	COI-2	Real, Reactive and Apparent Powers, Power Triangle, Power Factor	T BOOK [2], CH6,Page no 221-242R BOOK [2], CH13,Page no 510-511	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
9	CO1	COI-2	Solving Problems on DC and AC circuits	T BOOK [2], CH4,Page no 166-167, 169-174, 177-184	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
10	CO2	COI-3	Mesh Analysis: Procedure, Solving Numerical Problems using Mesh Analysis	R BOOK [1], CH4.3Page no 92-98	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
11	CO2	COI-3	Super Mesh Concept and Procedure, Solving Numerical Problems using Super Mesh Analysis	R BOOK [1], CH4.3 Page no 98-101	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
12	CO2	COI-3	Nodal Analysis: Procedure, Solving Numerical Problems using Nodal Analysis	R BOOK [1], CH4.1Page no 80-89	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
13	CO2	COI-3	Super Node Concept and Procedure, Solving Numerical Problems using Super Node Analysis	R BOOK [1], CH4.1Page no 89-91	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
14	CO2	COI-3	Superposition Theorem: Statement, Procedural Steps, Solving Numerical Problems using Superposition Theorem	T BOOK [1],CH35,Page no 518-527	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
15	CO2	COI-3	Thevenin's Theorem: Statement, Procedural Steps, Solving Numerical Problems using Thevenin's Theorem	T BOOK [1],CH36,Page no 528-539	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
16	CO2	COI-3	Norton's Theorem: Statement, Procedural	T BOOK	Chalk,PPT,Talk	ALM,End Semester Exam,Home

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			Steps, Solving Numerical Problems using Norton's Theorem	[1],CH36,Page no 539-551		Assignment,SEM-EXAM1
17	CO2	COI-3	Maximum Power Transfer Theorem: Statement, Procedural Steps, Solving Numerical Problems using Maximum Power Transfer Theorem	T BOOK [2], CH3,Page no 130-133	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
18	CO2	COI-2	Steady State Analysis of RL, RC and RLC Circuits	T BOOK [2],CH5,Page no 189-195	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
19	CO2	COI-3	Series Resonance - Condition for Resonance in Series RLC Circuit, Impedance and Current Characteristics of Resonant Circuit, Parameters of Resonant Circuits and Numerical Problems	T BOOK [2],CH8,Page no 284-298	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
20	CO3	COI-1	Conductors, Insulators and Semiconductors, Types of Semiconductors, Unbiased PN Junction	T BOOK [3],CH1Page no 5-11	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
21	CO3	COI-2	PN Diode: Operation, V-I Characteristics and Effect of Temperature, Applications of diode, Diode as a Switch	T BOOK [3],CH1, Page no 10-21	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
22	CO3	COI-2	Half Wave Rectifier: Operation, Ripple Factor, Efficiency, PIV and % Regulation, Numerical Problems	T BOOK [3],CH2,Page no 71-75	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
23	CO3	COI-2	Center-Tapped Full Wave Rectifier: Operation, Ripple Factor, Efficiency, PIV and % Regulation, Numerical Problems	T BOOK [3],CH2,Page no 75-79	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
24	CO3	COI-2	Bridge Full Wave Rectifier: Operation, PIV, Comparison of Rectifiers and Numerical Problems	T BOOK [3],CH2,Page no 75-78	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
25	CO3	COI-2	Clippers: Types, Transfer Characteristics, Clippers without Bias Voltage, Clippers with Bias Voltage	T BOOK [3],CH2,Page no 78-85	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
26	CO3	COI-2	Positive & Negative Clampers: Operation, Input & Output Waveforms	TBOOK [3],CH2Page no 85-88	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
27	CO3	COI-2	Zener Diode: Avalanche and Zener Breakdowns, Zener Diode V-I Characteristics, Zener as a Regulator	T BOOK [3],CH1,Page no 38-41	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
28	CO3	COI-2	LED: Working Principle, Specifications, Advantages and Applications	T BOOK [3],CH1,Page no 41-48	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
29	CO3	COI-2	Bipolar Junction Transistor: Operation, Input and Output Characteristics of CB Configuration, Base Width Modulation, Amplification Factor α	T BOOK [3],CH3,Page no 129-135	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
30	CO3	COI-2	Input and Output Characteristics of CE and CC Configurations with Amplification Factors β and γ , Relation between α , β and γ , Transistor as switch	T BOOK [3],CH3,Page no -136-145	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
31	CO4	COI-1	Introduction, Types and Advantages of ICs, Comparison between Analog and Digital ICs, Introduction to Op-Amp,	T BOOK [4],CH1,Page no -17-26	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			Virtual Short Circuit Concept			
32	CO4	COI-2	Ideal and Practical Characteristics of Op-Amp, Transfer Characteristics of Op-Amp, Pin Diagram of IC 741 Op-Amp	T BOOK [4],CH2,Page no -58-60	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
33	CO4	COI-2	Configurations of Op-Amp: Open Loop and Closed Loop, Inverting Amplifier & its Gain	T BOOK [4],CH2,6Page no -60-62, 216-217	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
34	CO4	COI-2	Non-Inverting Amplifier & its Gain, Applications of Op-Amp	T BOOK [4],CH6,Page no -218	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
35	CO4	COI-2	555 Timer: Pin Diagram, Internal Structure, Modes of Operation – Astable Mode	T BOOK [4],CH9,Page no -416-428	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
36	CO4	COI-2	Modes of Operation – Monostable and Bistable Modes, Features & Applications of 555 Timer	T BOOK [4],CH9,Page no -416-428	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
37	CO4	COI-2	Voltage Regulators: Introduction and Types, Pin Diagrams of 3-Terminal IC Voltage Regulators – 78XX, 79XX, Specifications and Applications of 78XX and 79XX	T BOOK [4],CH9,Page no -453-454	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
38	CO4	COI-2	LM723 IC Regulator – Pin Diagram, Specifications, Advantages & Disadvantages, Applications of LM723	T BOOK [4],CH9,Page no -457-461	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
39	CO4	COI-2	Revision and Solving Problems on Op-Amp, 555 Timer and Voltage Regulators	T BOOK [4],CH6,9Page no -213, 416-428	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Students gets the knowledge on the course rationale and evaluation methods

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Briefing on Course Rationale	1	PPT	--- NOT APPLICABLE ---
20	Briefing on Course Handout (CHO) and Evaluation methods	1	PPT	--- NOT APPLICABLE ---
5	Summary & Conclusion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 Students gets knowledge on basic circuit elements

Session Outcome: 2 Students are able to calculate current and voltage of a circuit using Ohm's law

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Basic Definitions, Passive Elements – R, L, and C	2	PPT	--- NOT APPLICABLE ---
20	Ideal and Practical Sources, Ohm's Law	2	PPT	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 Students are able to apply KVL and KCL to determine current and voltage in a circuit

Time(min)	Topic	BTL	Teaching-Learning	Active Learning
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			Methods	Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Kirchhoff's Current Law and Solving Problems	2	PPT	Problem-Based Learning
20	Kirchhoff's Voltage Law and Solving Problems	2	Chalk	Problem-Based Learning
5	Summary & Conclusion	2	Talk	Problem-Based Learning

SESSION NUMBER : 4

Session Outcome: 1 Students are able to optimize any electrical or electronic circuit using series-parallel technique and source transformation method

Session Outcome: 2 Students are able to calculate current and voltages in a circuit by applying CDR and VDR

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Series & Parallel Connections of R, L, C	2	PPT	--- NOT APPLICABLE ---
20	Current Division and Voltage Division Rules, Source Transformation	2	PPT	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 1 Students gets knowledge on Star-Delta transformation

Session Outcome: 2 Students are able to optimize complex electric circuits by using Star-Delta transformation

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Star-to-Delta Transformation and Solving Problems	2	PPT	--- NOT APPLICABLE ---
20	Delta-to-Star Transformation and Solving Problems	2	PPT	--- NOT APPLICABLE

5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 1 Students gets knowledge on fundamentals of AC signals

Session Outcome: 2 Students are able to calculate RMS value, Average Value, Form & Peak factors of different AC signals

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Basic Definitions of AC Signals, Average & RMS Values	2	PPT	One minute paper
20	Peak & Form Factors of Different AC Signals	2	PPT	One minute paper
5	Summary & Conclusion	2	Talk	One minute paper

SESSION NUMBER : 7

Session Outcome: 1 Students are able to calculate the reactance, impedance and phase in R, L, and C elements with AC source

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Analysis of R and L with AC Signal – Impedance, Reactance, Phase and Phase difference	2	PPT	--- NOT APPLICABLE ---
20	Analysis of C with AC Signal – Impedance, Reactance, Phase and Phase difference	2	PPT	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Students are able to calculate real, reactive and apparent powers and power factor of different AC circuits

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Real Power, Reactive Power and Apparent Power	2	PPT	--- NOT APPLICABLE ---
20	Power Triangle, Power Factor	2	PPT	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 Students are able to solve problems on DC and AC circuits using different techniques

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Solving Problems on DC circuits	2	PPT	--- NOT APPLICABLE ---
10	Solving Problems on AC circuits	2	PPT	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Students acquire knowledge about Mesh Analysis

Session Outcome: 2 Students are able to solve different electrical circuits using Mesh Analysis

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Mesh Analysis: Procedure	3	PPT	--- NOT APPLICABLE ---

30	Solving Numerical Problems using Mesh Analysis	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Students acquire knowledge about Super Mesh

Session Outcome: 2 Students are able to solve different electrical circuits using Super Mesh Concept

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Super Mesh Concept and Procedure	3	PPT	--- NOT APPLICABLE ---
30	Solving Numerical Problems using Super Mesh Analysis	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 Students acquire knowledge on Nodal Analysis

Session Outcome: 2 Students are able to solve different electrical circuits using Nodal Analysis

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Nodal Analysis: Procedure	3	PPT	Problem-Based Learning
30	Solving Numerical Problems using Nodal Analysis	3	Chalk	Problem-Based Learning
5	Summary & Conclusion	3	Talk	Problem-Based Learning

SESSION NUMBER : 13

Session Outcome: 1 Students acquire knowledge about Super Node

Session Outcome: 2 Students are able to solve different electrical circuits using Super Mesh Concept

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Super Node Concept and Procedure	3	PPT	--- NOT APPLICABLE ---
30	Solving Numerical Problems using Super Node Analysis	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 1 Students are able to solve different electrical circuits using Superposition Theorem

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Superposition Theorem: Statement, Procedural Steps	3	PPT	--- NOT APPLICABLE ---
30	Solving Numerical Problems using Superposition Theorem	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 1 Students are able to solve different electrical circuits using Thevenin's Theorem

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Thevenin's Theorem: Statement, Procedural Steps	3	PPT	Problem-Based Learning

30	Solving Numerical Problems using Thevenin's Theorem	3	Chalk	Problem-Based Learning
5	Summary & Conclusion	3	Talk	Problem-Based Learning

SESSION NUMBER : 16

Session Outcome: 1 Students are able to solve different electrical circuits using Norton's Theorem

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Norton's Theorem: Statement, Procedural Steps	3	PPT	--- NOT APPLICABLE ---
30	Solving Numerical Problems using Norton's Theorem	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 1 Students are able to solve different electrical circuits using Maximum Power Transfer Theorem

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Maximum Power Transfer Theorem: Statement, Procedural Steps	3	PPT	--- NOT APPLICABLE ---
30	Solving Numerical Problems using Maximum Power Transfer Theorem	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

Session Outcome: 1 Students gets knowledge on steady state analysis of RL, RC and RLC circuits with AC source

Session Outcome: 2 Students are able to calculate reactance and impedance in different RL, RC and RLC circuits with AC source

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Steady State Analysis of RL and RC Circuits	3	PPT	--- NOT APPLICABLE ---
10	Steady State Analysis of RLC Circuit	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

Session Outcome: 1 Students gets knowledge on resonance in electrical circuits comprising R, L, and C

Session Outcome: 2 Students are able to calculate different parameters of RLC series resonant circuit

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Series Resonance - Condition for Resonance in Series RLC Circuit, Impedance and Current Characteristics of Resonant Circuit	3	PPT	--- NOT APPLICABLE ---
20	Parameters of Resonant Circuits and Numerical Problems	3	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 1 Students gets knowledge on concepts of semiconductors and types

Session Outcome: 2 Students are able to understand the concept of barrier potential in an unbiased PN junction

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE

30	Conductors, Insulators and Semiconductors, Types of Semiconductors	2	PPT	--- NOT APPLICABLE ---
10	Unbiased PN Junction	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 1 Students are able to understand the working, V-I and thermal characteristics of a PN diode

Session Outcome: 2 Students gets knowledge on application of a diodes such as diode as a switch

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	PN Diode: Operation, V-I Characteristics and Effect of Temperature, Applications of diode	2	PPT	Video synthesis
10	Diode as a Switch	2	Chalk	Video synthesis
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

Session Outcome: 1 Students are able to understand the working of a Half Wave Rectifier and its different parameters

Session Outcome: 2 Students are able to solve problems on HWRs

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Half Wave Rectifier: Operation, Ripple Factor, Efficiency	2	PPT	--- NOT APPLICABLE ---
10	PIV and % Regulation, Numerical Problems	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE

SESSION NUMBER : 23

Session Outcome: 1 Students are able to understand the working of a Center-Tapped Full Wave Rectifier and its different parameters

Session Outcome: 2 Students are able to solve problems on Center-Tapped FWRs

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Center-Tapped Full Wave Rectifier: Operation, Ripple Factor, Efficiency,	2	PPT	--- NOT APPLICABLE ---
10	PIV and % Regulation, Numerical Problems	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

Session Outcome: 1 Students are able to understand the working of a Bridge Full Wave Rectifier and its different parameters

Session Outcome: 2 Students are able to solve problems on Bridge FWRs

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Bridge Full Wave Rectifier: Operation, PIV	2	PPT	Brain storming session
20	Comparison of Rectifiers and Numerical Problems	2	Chalk	Brain storming session
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 25

Session Outcome: 1 Students gets knowledge on different wave shaping circuits

Session Outcome: 2 Students are able to the applications of diode as a clipper

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Clippers: Types, Transfer Characteristics, Clippers without Bias Voltage	2	PPT	--- NOT APPLICABLE ---
10	Clippers with Bias Voltage	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 26

Session Outcome: 1 Students gets knowledge on different dc level shifters

Session Outcome: 2 Students are able to know the applications of diode as a clamper

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Positive Clampers: Operation, Input & Output Waveforms	2	PPT	--- NOT APPLICABLE ---
20	Negative Clampers: Operation, Input & Output Waveforms	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 27

Session Outcome: 1 Students are able to understand the difference between avalanche and Zener breakdowns, working of Zener diode

Session Outcome: 2 Students gets knowledge on how Zener diode working as a voltage regulator

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---

10	Zener Diode: Avalanche and Zener Breakdowns	2	PPT	--- NOT APPLICABLE ---
30	Zener Diode V-I Characteristics, Zener as a Regulator	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 28

Session Outcome: 1 Students are able to understand the working of LED

Session Outcome: 2 Students gets knowledge on advantages and application of LED

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	LED: Working Principle, Specifications	2	PPT	--- NOT APPLICABLE ---
10	LED Advantages and Applications	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 29

Session Outcome: 1 Students are able to understand the working of BJT

Session Outcome: 2 Students gets knowledge on input & output characteristics of CB configuration and on base width modulation

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Bipolar Junction Transistor: Operation, Input and Output Characteristics of CB Configuration	2	PPT	--- NOT APPLICABLE ---
10	Base Width Modulation, Amplification Factor α	2	Chalk	--- NOT APPLICABLE ---

5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---
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SESSION NUMBER : 30

Session Outcome: 1 Students are able to understand the input & output characteristics of CE and CC configuration

Session Outcome: 2 Students gets knowledge on working of transistor as a switch

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Input and Output Characteristics of CE Configurations with Amplification Factors β	2	PPT	--- NOT APPLICABLE ---
20	Input and Output Characteristics of CC Configurations with Amplification γ , Relation between α , β and γ , Transistor as switch	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 31

Session Outcome: 1 Students gets knowledge on analog and digital ICs and their advantages and disadvantages

Session Outcome: 2 Students are able to understand the concept of virtual short circuit in an Op-Amp

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Introduction, Types and Advantages of ICs, Comparison between Analog and Digital ICs	2	PPT	--- NOT APPLICABLE ---
20	Introduction to Op-Amp, Virtual Short Circuit Concept	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 32

Session Outcome: 1 Students gets knowledge on ideal and practical characteristics of Op-Amp

Session Outcome: 2 Students are able to understand the pin configuration of IC 741 Op-Amp

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Ideal and Practical Characteristics of Op-Amp, Transfer Characteristics of Op-Amp	2	PPT	--- NOT APPLICABLE ---
10	Pin Diagram of IC 741 Op-Amp	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 33

Session Outcome: 1 Students gets knowledge on open and closed loop configurations of Op-Amp

Session Outcome: 2 Students are able to understand the operation of inverting amplifier and derive its voltage gain

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Configurations of Op-Amp: Open Loop and Closed Loop	2	PPT	Design Thinking
20	Inverting Amplifier & its Gain	2	Chalk	Design Thinking
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 34

Session Outcome: 1 Students are able to understand the operation of non-inverting amplifier and derive its voltage gain

Session Outcome: 2 Students are able to list the applications of Op-Amp

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---

20	Non-Inverting Amplifier & its Gain	2	PPT	--- NOT APPLICABLE ---
20	Applications of Op-Amp	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 35

Session Outcome: 1 Students gets knowledge on the basics on IC 555 timer

Session Outcome: 2 Students are able to understand different modes of operation of IC 555 timer

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	555 Timer: Pin Diagram, Internal Structure	2	PPT	--- NOT APPLICABLE ---
20	Modes of Operation – Astable Mode	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 36

Session Outcome: 1 Students are able to understand monostable and bistable modes of IC 555 timer

Session Outcome: 2 Students are able to list the application of IC 555 timer

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Modes of Operation – Monostable and Bistable Modes	2	PPT	Sketching & Drawing
10	Features & Applications of 555 Timer	2	Chalk	Sketching & Drawing
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 37

Session Outcome: 1 Students gets knowledge on different IC voltage regulators

Session Outcome: 2 Students are able to list the application of IC 78XX and 79XX voltage regulators

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
30	Voltage Regulators: Introduction and Types, Pin Diagrams of 3-Terminal IC Voltage Regulators – 78XX, 79XX	2	PPT	--- NOT APPLICABLE ---
10	Specifications and Applications of 78XX and 79XX	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 38

Session Outcome: 1 Students gets knowledge on IC 723 voltage regulator

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	LM723 IC Regulator – Pin Diagram, Specifications	2	PPT	--- NOT APPLICABLE ---
20	Advantages & Disadvantages, Applications of LM723	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 39

Session Outcome: 1 Students recall and summarize the concepts, types, and applications of all ICs including 741 Op-Amp, 555 Timer, and voltage regulators

Session Outcome: 2 Students are able to solve numerical problems on Op-Amp and 555 Timers

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Revision on IC 741 and Solving Problems	2	PPT	--- NOT APPLICABLE ---
20	Revision 555 Timer, IC Regulators and Solving Problems	2	Chalk	--- NOT APPLICABLE ---
5	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan: NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Demonstration of Different Electrical & Electronic Components	CO5
2	Introduction of NI Multisim for Circuit Analysis	CO5
3	Verification of Ohm's Law, KVL and KCL	CO5
4	Verification of Superposition Theorem	CO5
5	Verification of Maximum Power Transfer Theorem	CO5
6	Verification of Resonance in Series RLC Circuit	CO5
7	Diode Rectifiers	CO5
8	Voltage Limiters	CO5
9	DC Level Shifters	CO5
10	Transistor as a Switch	CO5
11	Inverting Amplifier using IC 741	CO5
12	Non-Inverting Amplifier using IC 741	CO5

Tutorial Session no	Topics	CO-Mapping
13	Astable Multivibrator using IC 555 Timer	CO5
14	Power Supply using IC Voltage Regulators	CO5
15	Star-Delta Transformation	CO5
16	Diode Rectifiers with Filters	CO5
17	Traffic Light System using IC 555 Timer	CO5

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Students able to identify and know the usage of different circuit components and equipment's and calculate the resistance value using color code

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction/Recap	2	PPT	--- NOT APPLICABLE ---
40	Introduction to different electrical and electronic components	2	PPT	--- NOT APPLICABLE ---
40	Identifying the components and their usage	2	PPT	--- NOT APPLICABLE ---
10	Summary & Intraction	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 Students able to simulate basic electrical and electronic circuits using NI Multisim software for better visualization and analysis

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Introducing NI Multisim tool	2	PPT	--- NOT APPLICABLE

20	Simple circuit demonstration in Multisim and Viva Voce	2	PPT	--- NOT APPLICABLE ---
10	Summary & Conclusion	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 Students able to verify Ohm's Law, Kirchhoff's Voltage Law (KVL), and Kirchhoff's Current Law (KCL) through experimental circuit setup

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Theoretical concepts of Ohm's Law, KVL, and KCL	2	PPT	--- NOT APPLICABLE ---
35	Practical verification of Ohm's Law, KVL, and KCL	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 1 Students able to validate the Superposition Theorem using linear electrical networks and measure the resulting voltages/currents

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Explanation of Superposition Theorem	2	PPT	--- NOT APPLICABLE ---
35	Experimental verification using a linear circuit	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE

10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 1 Students able to analyze and confirm the conditions for maximum power transfer in resistive circuits.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Understanding Maximum Power Transfer Theorem	2	PPT	--- NOT APPLICABLE ---
35	Practical demonstration with varying load resistances	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 1 Students able to analyze the behavior of series RLC circuit and determine the resonance frequency.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Concept of resonance in RLC circuits	2	PPT	--- NOT APPLICABLE ---
35	Practical setup and observation of resonance condition	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE

SESSION NUMBER : 7

Session Outcome: 1 Students able to construct and test the operation of half-wave and full-wave rectifier circuits using diodes

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Introduction to rectifiers and circuit design	2	PPT	--- NOT APPLICABLE ---
35	Building and testing halfwave and fullwave rectifiers	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Students able to design and verify voltage clipping circuits using diodes to limit signal amplitudes.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Theory of clippers and their applications	2	PPT	--- NOT APPLICABLE ---
35	Designing and testing various clipping circuits	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 Students able to implement and analyze clamping circuits to shift the DC level of AC signals.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Principle of DC level shifting and clamper circuits	2	PPT	--- NOT APPLICABLE ---
35	Practical implementation of clampers using diodes	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Students able to demonstrate the working of a BJT transistor as an electronic switch in circuit applications.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Introduction to BJT and its operation as a switch	2	PPT	--- NOT APPLICABLE ---
35	Building and testing transistor switch circuits	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Students able to design and test an inverting amplifier configuration using operational amplifier IC 741.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Overview of op-amp characteristics and inverting mode	2	PPT	--- NOT APPLICABLE ---
35	Designing and testing inverting amplifier using IC 741	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 Students able to design and test a non-inverting amplifier configuration using operational amplifier IC 741.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Understanding non-inverting amplifier configuration	2	PPT	--- NOT APPLICABLE ---
35	Building and testing non-inverting amplifier using IC 741	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 1 Students able to construct an astable multivibrator circuit using IC 555 to generate continuous square wave pulses.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Basics of IC 555 timer and astable mode	2	PPT	--- NOT APPLICABLE ---
35	Constructing and analyzing astable multivibrator	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 1 Students able to design a regulated DC power supply circuit using IC voltage regulators (e.g., 7805/7905)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Concept of voltage regulation using ICs	2	PPT	--- NOT APPLICABLE ---
35	Designing a power supply using IC 78XX regulators	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 1 Students able to convert and analyze resistive electrical networks using star-delta and delta-star transformations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---

25	Theory of star-delta and delta-star transformation	2	PPT	--- NOT APPLICABLE ---
35	Applying transformations to solve network problems	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

Session Outcome: 1 Students able to implement filtering techniques in diode rectifier circuits to reduce ripple and obtain smooth DC output

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Review of rectifier output and ripple factor	2	PPT	--- NOT APPLICABLE ---
35	Adding filters to rectifiers and observing effects practically	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 1 Students able to design and simulate a simple traffic light control system using IC 555 timer and LEDs

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
25	Logic behind traffic light systems and timer basics	2	PPT	--- NOT APPLICABLE ---

35	Building and simulating traffic light control circuit using IC 555	4	PPT	--- NOT APPLICABLE ---
20	Results & Viva-voce	4	Talk	--- NOT APPLICABLE ---
10	Summary & Conclusion	4	Talk	--- NOT APPLICABLE ---

Skilling Course DELIVERY Plan: NO Delivery Plan Exists

Skilling Session wise Teaching – Learning Plan

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
3	Weekly Homework Assignments	1	AC Fundamentals	Average & RMS values, Peak and Form Factors of different AC Signals, Analysis of R, L and C with AC Signal	CO1
6	Weekly Homework Assignments	2	Steady State Analysis of RL, RC and RLC Circuits	Steady State Analysis of RL, RC and RLC Circuits, Resonant Circuits	CO2
9	Weekly Homework Assignments	3	Diode Application, Zener Diode, LED and BJT	Rectifiers, Clippers, Clampers, LED, and Transistor Characteristics	CO3
12	Weekly Homework Assignments	4	Analog ICs	IC 741, 555 Timer and Voltage Regulators	CO4

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	- - -	---	V-S9,V-S10	V-S9,V-S10,V-S45,V-S46,V-S47,V-S48	V-S17,V-S18	V-S17,V-S18	---	---	V-S33,V-S34
	Tutorial	- - -	---	--	--	--	--	---	---	--

	Lab	- - -	---	V-S11,V-S11,V-S11,V-S12,V-S12,V-S12	V-S11,V-S11,V-S11,V-S12,V-S12	V-S19,V-S19,V-S19,V-S20,V-S20,V-S20	V-S19,V-S19,V-S19,V-S20,V-S20	---	---	V-S35,V-S35,V-S35,V-S36,V-S36,V-S36
	Skilling	- - -	---	--	--	--	--	---	---	--
Tue	Theory	- - -	V-S13,V-S14,V-S15,V-S16	V-S11,V-S12	V-S1,V-S2,V-S3,V-S4,V-S11,V-S12,V-S37,V-S38,V-S39,V-S40	V-S19,V-S20	V-S19,V-S20	---	---	V-S35,V-S36
	Tutorial	- - -	--	--	--	--	--	---	---	--
	Lab	- - -	--	V-S9,V-S9,V-S9,V-S10,V-S10,V-S10	V-S9,V-S9,V-S9,V-S10,V-S10	V-S17,V-S17,V-S17,V-S18,V-S18,V-S18	V-S17,V-S17,V-S17,V-S18,V-S18	---	---	V-S33,V-S33,V-S33,V-S34,V-S34,V-S34
	Skilling	- - -	--	--	--	--	--	---	---	--
Wed	Theory	- - -	---	V-S1,V-S2	V-S1,V-S2,V-S41,V-S42,V-S43,V-S44	V-S21,V-S22	V-S5,V-S6,V-S7,V-S8,V-S21,V-S22	V-S45,V-S46	V-S45,V-S46	V-S37,V-S38
	Tutorial	- - -	---	--	--	--	--	--	--	--
	Lab	- - -	---	V-S3,V-S3,V-S3,V-S4,V-S4,V-S4	V-S3,V-S3,V-S3,V-S4,V-S4	V-S23,V-S23,V-S23,V-S24,V-S24,V-S24	V-S23,V-S23,V-S23,V-S24,V-S24	V-S47,V-S47,V-S47,V-S48,V-S48,V-S48	V-S47,V-S47,V-S47,V-S48,V-S48	V-S39,V-S39,V-S39,V-S40,V-S40,V-S40
	Skilling	- - -	---	--	--	--	--	--	--	--
Thu	Theory	- - -	V-S17,V-S18,V-S19,V-S20,V-	V-S3,V-S4	V-S3,V-S4	V-S23,V-S24	V-S23,V-S24	V-S47,V-S48	V-S47,V-S48	V-S39,V-S40

			S49,V-S50,V-S51,V-S52							
	Tutorial	- - -	--	--	--	--	--	--	--	--
	Lab	- - -	--	V-S1,V-S1,V-S1,V-S2,V-S2,V-S2	V-S1,V-S1,V-S1,V-S2,V-S2,V-S2	V-S21,V-S21,V-S21,V-S22,V-S22,V-S22	V-S21,V-S21,V-S21,V-S22,V-S22,V-S22	V-S45,V-S45,V-S45,V-S46,V-S46,V-S46	V-S45,V-S45,V-S45,V-S46,V-S46,V-S46	V-S37,V-S37,V-S37,V-S38,V-S38,V-S38
	Skilling	- - -	--	--	--	--	--	--	--	--
Fri	Theory	- - -	V-S21,V-S22,V-S23,V-S24	V-S5,V-S6	V-S5,V-S6,V-S33,V-S34,V-S35,V-S36	V-S13,V-S14	V-S9,V-S10,V-S11,V-S12,V-S13,V-S14	V-S49,V-S50	V-S49,V-S50	V-S41,V-S42
	Tutorial	- - -	--	--	--	--	--	--	--	--
	Lab	- - -	--	V-S7,V-S7,V-S7,V-S8,V-S8,V-S8	V-S7,V-S7,V-S7,V-S8,V-S8,V-S8	V-S15,V-S15,V-S15,V-S16,V-S16,V-S16	V-S15,V-S15,V-S15,V-S16,V-S16,V-S16	V-S51,V-S51,V-S51,V-S52,V-S52,V-S52	V-S51,V-S51,V-S51,V-S52,V-S52,V-S52	V-S43,V-S43,V-S43,V-S44,V-S44,V-S44
	Skilling	- - -	--	--	--	--	--	--	--	--
Sat	Theory	- - -	---	V-S7,V-S8	V-S7,V-S8	V-S15,V-S16	V-S15,V-S16	V-S51,V-S52	V-S51,V-S52	V-S43,V-S44
	Tutorial	- - -	---	--	--	--	--	--	--	--
	Lab	- - -	---	V-S5,V-S5,V-S5,V-S6,V-S6,V-S6	V-S5,V-S5,V-S5,V-S6,V-S6,V-S6	V-S13,V-S13,V-S13,V-S14,V-S14,V-S14	V-S13,V-S13,V-S13,V-S14,V-S14,V-S14	V-S49,V-S49,V-S49,V-S50,V-S50,V-S50	V-S49,V-S49,V-S49,V-S50,V-S50,V-S50	V-S41,V-S41,V-S41,V-S42,V-S42,V-S42
	Skilling	- - -	---	--	--	--	--	--	--	--
Sun	Theory	- -	--	--	--	--	--	--	--	--

	Tutorial	-	--	--	--	--	--	--	--	--
	Lab	-	--	--	--	--	--	--	--	--
	Skilling	-	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4	CO5
End Semester Summative Evaluation Total= 40 %	Lab End Semester Exam	Weightage	16		100					16
		Max Marks	50							50
	End Semester Exam	Weightage	24		180	6	6	6	6	
		Max Marks	100			25	25	25	25	
In Semester Formative Evaluation Total= 24 %	Continuous Evaluation - Lab Exercise	Weightage	10		100					10
		Max Marks	50							50
	Home Assignment and Textbook	Weightage	6		100	1.5	1.5	1.5	1.5	
		Max Marks	100			25	25	25	25	
	ALM	Weightage	8		100	2	2	2	2	
		Max Marks	100			25	25	25	25	
In Semester Summative Evaluation Total= 36 %	Lab In Semester Exam	Weightage	8		100					8
		Max Marks	50							50
	Semester in Exam-II	Weightage	14		90			7	7	
		Max Marks	50					25	25	

	Semester in Exam-I	Weightage	14			7	7			
		Max Marks	50		90	25	25			

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course. In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments.

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Veera Venkata Kameswara Srinivas Prasad Mutya	L	5-MA,10-MA	-	-	-	-
Veera Venkata Kameswara Srinivas Prasad Mutya	P	5-A,10-A	-	-	-	-
Suresh Namgiri	L	1-MA,33-MA,49-MA	-	-	-	-
Suresh Namgiri	P	1-A,33-A,49-A	-	-	-	-
Somlal Jarupula	L	12-MA,44-MA	-	-	-	-
Kali prasad	L	2-MA,34-MA	-	-	-	-
Kali prasad	P	2-A,6-B,13-B,19-B,34-A,43-C	-	-	-	-

Gnana Maddula	L	24-MA	-	-	-	-
Gnana Maddula	P	24-A	-	-	-	-
Venkata Durga Mareedu	P	7-C	-	-	-	-
Siddaiah Nalluri	P	21-C,48-C	-	-	-	-
Venkateswara Rao Alaparthi	P	4-B,7-B	-	-	-	-
Babu Shaik	P	9-B	-	-	-	-
Mahamuda Shaik	P	3-B	-	-	-	-
Sudarsa Dora Babu	P	14-C	-	-	-	-
Katta Rajesh Babu	L	35-MA,45-MA,50-MA	-	-	-	-
Katta Rajesh Babu	P	10-C,35-A,45-A,50-A	-	-	-	-
KONERU SWAPNA	L	46-MA	-	-	-	-
KONERU SWAPNA	P	46-A	-	-	-	-
Agilesh R	P	2-C,52-B	-	-	-	-
MOHAN K.N	P	2-B	-	-	-	-
Raghavendra Kumar Kanike	P	8-B	-	-	-	-
Aravindan Srinivasan	P	16-C,34-C	-	-	-	-
Dasari Manasa	P	4-C	-	-	-	-
Arumalla Raju	P	22-C,23-C,35-C	-	-	-	-
Ushasree Linginedi	P	12-C,15-C,49-C	-	-	-	-
Inakollu Aswani	P	3-C,5-C	-	-	-	-
Dharmendra Pilli	P	11-B,13-C,52-C	-	-	-	-
Pallapati Kumar	P	50-C	-	-	-	-
NAGESH MANTRAVADI	L	3-MA,21-MA	-	-	-	-
NAGESH MANTRAVADI	P	3-A,21-A,35-B	-	-	-	-
SIVANI PINNABOINA	P	24-B,39-B,45-C	-	-	-	-
NAGAMALLI ARASAVALLI	P	44-C	-	-	-	-
CHINTA MANJUSHA	P	36-B,40-C	-	-	-	-

SIVAPRASAD LEBAKA	L	14- MA,42- MA	-	-	-	-
SIVAPRASAD LEBAKA	P	12-A,14- A,20- B,38- B,39- C,42-A	-	-	-	-
TAMMINENI SREELATHA	L	15- MA,20- MA	-	-	-	-
TAMMINENI SREELATHA	P	15-A,20- A,40- B,42- B,44- B,45-B	-	-	-	-
BANDARU MAMATHA	P	9-C	-	-	-	-
SRINIVASARAO ALLURI	L	6-MA,16- MA,38- MA	-	-	-	-
SRINIVASARAO ALLURI	P	6-A,16- A,38- A,41-B	-	-	-	-
RAJANIDEVI MERIGALA	L	17- MA,47- MA	-	-	-	-
RAJANIDEVI MERIGALA	P	1-B,16- B,17- A,44- A,47-A	-	-	-	-
LAKSHMUNAIDU M	L	8-MA,18- MA,40- MA	-	-	-	-
LAKSHMUNAIDU M	P	8-A,15- B,18- A,40-A	-	-	-	-
RANJAN MAHAPATRA	P	49-B	-	-	-	-
Anka Siva Sonti	P	19-C,21- B,37-C	-	-	-	-
Vineetha K V	P	18-C,51- B	-	-	-	-
Hushnud Hushnud	P	10-B	-	-	-	-
GANESH BABU RAJENDRAN	L	9-MA,36- MA	-	-	-	-
GANESH BABU RAJENDRAN	P	9-A,36- A,48- B,50-B	-	-	-	-

Anand Dhanakotti	P	8-C	-	-	-	-
HARI CHANDANA PICHHIKA	L	37- MA,43- MA	-	-	-	-
HARI CHANDANA PICHHIKA	P	1-C,5- B,12- B,33- C,37- A,43-A	-	-	-	-
Keisham Pritamdas	L	22- MA,39- MA	-	-	-	-
Keisham Pritamdas	P	22-A,36- C,39- A,42- C,51-C	-	-	-	-
CHANDAN KUMAR	L	11- MA,51- MA	-	-	-	-
CHANDAN KUMAR	P	11-A,22- B,23- B,33- B,46- B,51-A	-	-	-	-
DEVADAS KUNA	P	46-C,47- C	-	-	-	-
Harsha Sri Kanakavalli	L	13- MA,48- MA	-	-	-	-
Harsha Sri Kanakavalli	P	13-A,18- B,37- B,41- C,43- B,48-A	-	-	-	-
NEPPALLI RAMESH	L	4-MA,52- MA	-	-	-	-
NEPPALLI RAMESH	P	4-A,11- C,47- B,52-A	-	-	-	-
Raktim Guha	L	7-MA,41- MA	-	-	-	-
Raktim Guha	P	7-A,17- C,20- C,38- C,41-A	-	-	-	-
Sukanta Swain	L	19- MA,23- MA	-	-	-	-
Sukanta Swain	P	6-C,19- A,23-	-	-	-	-

		A,34-B				
Kaja Mohan	P	14-B,17-B,24-C	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(SRINIVASARAO ALLURI)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of DBES-2

HEAD OF DEPARTMENT:**Approval from: DEAN-ACADEMICS**

(Sign with Office Seal) [object HTMLDivElement]